Perceval Beja-Battais

PhD Student in Applied Mathematics

Centre Borelli, ENS Paris-Saclay · in collaboration with Framatome

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Imitation Learning · Dynamical Systems · Machine Learning · Model Predictive Control

EDUCATION

PhD in Applied Mathematics Apr 2024 – Mar 2027
Centre Borelli, ENS Paris-Saclay & Framatome — supervised by Nicolas Vayatis · Machine Learning for Optimal Control, Machine Learning for Simulation, Sensitivity Analysis, Physics-Informed ML, Safe ML — with application to the nuclear industry.

MVA Master (Mathematics, Vision, Learning) — Highest Honours 2022 – 2023
ENS Paris-Saclay · Statistical Learning, Deep Learning, Machine Learning, Computer Vision.

Mathematics of Modeling Master 2021 – 2022
Sorbonne Université · Optimization, Optimal Transport, Functional Analysis, Game Theory, Statistics, PDE.

Ingénieur Civil des Mines 2019 – 2022
Mines Nancy · Probabilities, Statistics, PDE, Optimization, Machine Learning and Deep Learning.

MPSI – MP 2017 – 2019
Lycée Lakanal · Two intensive years preparing the entrance exams of top French engineering schools.

PUBLICATIONS

P. Beja-Battais, A. Grossetête, N. Vayatis. **Structured Representation Learning for Behavior Cloning: How can we learn to safely control a nuclear power plant?** *NeurIPS 2026 Workshop AI4PowerGrids* arXiv:2610.06211

P. Beja-Battais, G. Dupré, A. Grossetête, N. Vayatis. **Accelerating Full-Scale Nonlinear Model Predictive Control via Surrogate Dynamics Optimization.** *Preprint, 2026* arXiv:2604.05566

P. Beja-Battais, A. Grossetête, N. Vayatis. **Enhancing Nuclear Reactor Core Simulation through Data-Based Surrogate Models.** *International Congress on Advances in Nuclear Power Plants (ICAPP), 2025* hal-05360255

G. Serré, P. Beja-Battais, S. Chirrane, A. Kalogeratos, N. Vayatis. **LIPO+: Frugal Global Optimization for Lipschitz Functions.** *SETN 2024* arXiv:2406.19723

P. Beja-Battais. **Overview of AdaBoost: Reconciling its views to better understand its dynamics.** *Preprint, 2023* arXiv:2310.18323

Patent

P. Beja-Battais, A. Grossetête, N. Vayatis. **A novel warm-start method for Optimal Control of Nuclear Power Plants.** *Patent, 2025*

RESEARCH & WORK EXPERIENCE

Pre-Doctoral Researcher Apr 2023 – Mar 2024
Centre Borelli, ENS Paris-Saclay · Research on Machine Learning topics.

R&D Intern Apr – Aug 2022
Framatome, Paris La Défense · Implementation from scratch of an optimization solver inside a dynamical system controlled by MPC algorithms.

Data Analyst Intern Jun – Aug 2021
Editec UK Ltd., London (UK) · Modeling & forecasting of lotto sales; analysis of players' behaviour to design targeted new games.

TALKS, POSTERS & AWARDS

Poster — Machine Learning for Optimal Control of Nuclear Power Plants Sep 2026
Machine Learning Summer School (MLSS), Max Planck Institute for Intelligent Systems, Tübingen

Talk — Enhancing Nuclear Reactor Core Simulation through Data-Based Surrogate Models 2025
International Congress on Advances in Nuclear Power Plants (ICAPP)

1st Prize — RL Challenge Accenta 2024
Collège de France

TEACHING & SUPERVISION

Teaching Assistant — Introduction to Statistical Learning 2026 – 2027
MVA Master, ENS Paris-Saclay

Mock oral examiner for engineering school entrance exams 2021 – 2025
IPESUP (MP, 2025) · Lycée Janson-de-Sailly (BCPST-2A, 2022–2023) · Lycée Condorcet (MPSI, 2021–2023)

Research internship supervision 2026
Efficient Sampling of Trajectories for Online Finetuning of Surrogate Simulation Schemes (May – Oct 2026)
Singular Perturbation of Nonlinear DAEs (Mar – Jun 2026)